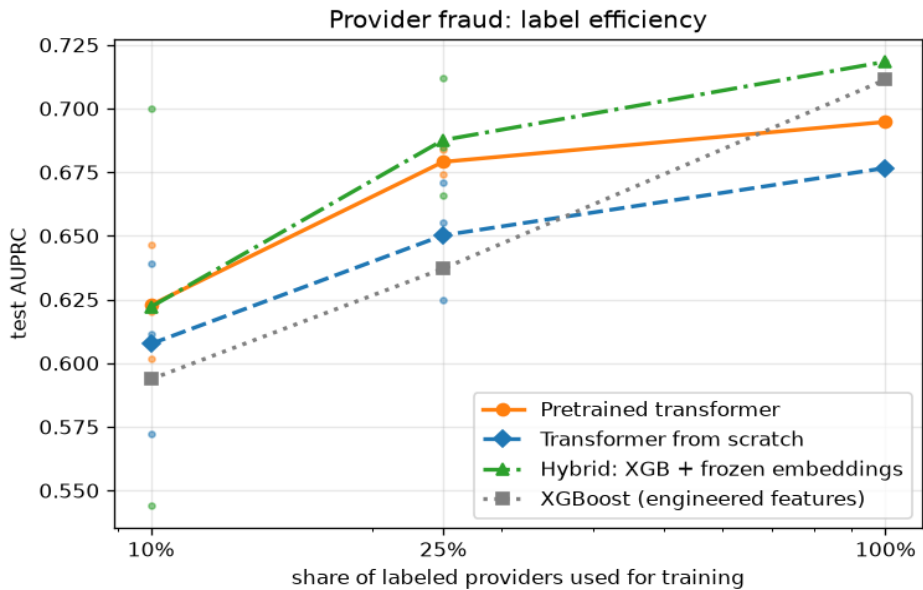


# claims-fm — a \$1.14 claims foundation model, benchmarked honestly

Pretrained a 17M-parameter claims encoder on 517k synthetic Medicare members (CMS DE-SynPUF, masked-code modeling); fine-tuned for the two problems a payer pays for — care-management risk stratification and provider fraud — against tuned XGBoost baselines frozen *before* any transformer training. Nicholi Caron · private repo `claims-fm` (access on request).

## Headline results (held-out test, single pass, bootstrap CIs in repo)

| Task                                      | Winner                           | Key numbers                                                                                       |
|-------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------|
| Member risk — top-decile cost (prev 10%)  | XGBoost                          | AUROC 0.762 vs transformer 0.715; capture@5% outreach 20.5%; post-isotonic ECE $\leq 0.007$       |
| Provider fraud — 100% labels (prev 9.4%)  | Hybrid: XGB + encoder embeddings | AUPRC 0.718, best precision@50 = 82%; plain LR 0.749 (reported straight)                          |
| Provider fraud — 25% of labels            | Pretrained encoder               | AUPRC 0.679 ( $\pm 0.004$ ) vs XGBoost 0.637 ( $\pm 0.050$ ) — better and 10 $\times$ more stable |
| Transfer ablation (identical arch/budget) | Pretrained > scratch             | From-scratch loses at every label fraction, on a dataset the encoder never saw                    |



The money chart: with 10–25% of fraud labels — the regime real SIUs operate in — pretraining beats both its from-scratch control and tuned XGBoost.

## What this demonstrates

- **Judgment over hype:** tuned XGBoost wins member risk on synthetic data and we ship it — then a hybrid ablation *measures* why (encoder embeddings add zero to engineered features here: the synthesis destroys sequence signal, not the method).
- **Label leverage:** confirmed fraud labels are a payer's scarcest resource; at 25% of labels the pretrained encoder approaches XGBoost's full-label performance, with 10 $\times$  lower variance.
- **Production discipline:** baselines frozen first; leakage rules as a 43-test suite; content-hashed splits; test sets scored once; calibration first-class; every dollar ledgered (\$1.14 of \$20).
- **Payment-integrity bridge:** provider billing anomalies are anomalous event streams — the same detection posture as security telemetry (my background): rank deviations, respect SIU caseload capacity.

Limitations stated plainly: synthetic data caps absolute numbers (ICD-9 era; constructed fraud labels; 512-token truncation). With real claims: ICD-10 vocab swap via GEMs, next-visit objective, monthly batch scoring with recalibration and drift monitors — and the hybrid test as the first cheap experiment.